Notes: Herkenhoff, Lise, Menzio, and Phillips (Econometrica, 2024)

Production and Learning in Teams

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1 Big picture

- **Question.** How much does a worker's human capital growth depend on the quality of his coworkers?
- **Core contribution.** The paper builds and estimates a search-theoretic labor-market model in which both production and human capital accumulation depend on coworkers' human capital.
- **Main empirical object.** The key reduced-form relationship is between a worker's wage in a new job after an employment-unemployment-employment transition and the average wage of his old coworkers, controlling for his old wage.
- **Why a model is needed.** Coworker wages are not randomly assigned. Since sorting is positive, coworkers' wages forecast a worker's own latent human capital. The model translates wage regressions into structural learning parameters.
- **Main empirical fact.** For workers who initially earn less than coworkers, a 10 percent increase in coworkers' average wage predicts a 1.23 percent higher wage in the next job. For workers who earn more than coworkers, the analogous effect is much weaker.
- **Learning result.** The estimated human capital accumulation function is convex in coworker human capital: workers learn from more knowledgeable coworkers, but are not meaningfully dragged down by less knowledgeable coworkers.
- **Production result.** The estimated production function is supermodular, though only mildly. Higher-human-capital workers are more productive when paired with higher-human-capital coworkers.
- **Sorting result.** The equilibrium does not display pure positive assortative matching or pure negative assortative matching. Workers are disproportionately matched with coworkers who have either slightly more or slightly less human capital.
- **Quantitative result.** Learning from coworkers accounts for about two thirds of the stock of human capital accumulated on the job. Learning by doing accounts for the remaining third.
- **Counterfactual result.** If production becomes more supermodular, labor-market segregation rises. Low-human-capital workers then have fewer chances to learn from better coworkers, reducing aggregate human capital and output.
- **Welfare result.** The equilibrium is inefficient because search bargaining wedges distort hiring and firing decisions. Quantitatively, the inefficiency is small: the planner has about 1 percent more human capital and 0.5 percent more output.
- **Takeaway.** Coworkers are not only contemporaneous production partners; they are inputs into future human capital. Sorting affects both current output and future skill accumulation.

2 Data and measurement

2.1 LEHD matched employer-employee data

- The paper uses the Longitudinal Employer-Household Dynamics data from 1998-2014 across 24 U.S. states.
- The data cover roughly 95 percent of U.S. private-sector jobs and contain workers, employers, quarterly earnings, and demographic characteristics.
- The data are linked to the 2000 Census, which provides education and occupation for a subset of workers.
- A worker's "wage" is annual labor earnings from the primary employer, measured as a log wage in the regressions.
- A worker's stable coworkers at firm j in year t are employees of the same firm who are full-year employees in t and have positive earnings from the firm in $t-1$ and $t+1$.

Interpretation. The empirical analysis uses wages as a noisy proxy for human capital. The structural model is needed because wages reflect human capital, bargaining, firm match value, and sorting.

2.2 E sample and EUE sample

- The broad E data set includes workers aged 24-65 who are fully employed, have Census demographic information, work at a single-unit firm, and have between 1 and 100 stable coworkers.
- The single-unit and coworker-size restrictions are designed to make it more likely that workers physically interact with the measured coworkers.
- The main EUE sample requires a worker to be employed in year t, experience unemployment in year $t+1$, and then be employed at a different firm in year $t+2$.

Interpretation. The EUE design is central. After a worker is hired from unemployment by a new firm, his wage is more likely to reflect accumulated human capital rather than the continuation value of the old job.

2.3 Outcome and treatment variables

- $w_{i,t+2}$: worker i's log wage in the new job after the EUE transition.
- $w_{i,t}$: worker i's log wage in the old job.
- $w^*_{j,t}$: average log wage of worker i's stable coworkers at old firm j.
- Workers are split into two groups:

$$w_{i,t} < w^*_{j,t} \quad \text{and} \quad w_{i,t} \geq w^*_{j,t}.$$

Interpretation. The split allows learning from better coworkers to differ from exposure to worse coworkers. The central finding is that the first channel is strong and the second is weak.

2.4 Additional heterogeneity variables

- **Education:** less than high school, high school, some college, college or more.
- **Occupation:** Abstract, Manual, and Routine categories based on Autor-Dorn task measures.
- **Age:** deciles of the worker's age distribution.
- **Coworker wage distribution:** mean, standard deviation, minimum, and maximum among stable coworkers.

Interpretation. These variables test whether the reduced-form coworker effect behaves like true learning. It should be stronger for workers who are more able to learn, jobs with more learning scope, and younger workers.

3 Models and empirical specifications

3.1 Individual wage regression: measuring coworker learning

The paper estimates

$$w_{i,t+2} = \phi_0 + \phi_1 w_{i,t} + \phi_2 w^*_{j,t} + \Gamma X_{i,j,t} + \varepsilon_{i,t}, \tag{7}$$

separately for workers with $w_{i,t} < w^*_{j,t}$ and $w_{i,t} \geq w^*_{j,t}$.

- $w_{i,t+2}$ is the post-transition log wage.
- $w_{i,t}$ is the worker's pre-transition log wage.
- $w^*_{j,t}$ is the pre-transition average coworker log wage.
- $X_{i,j,t}$ includes state, 1-digit SIC, race, gender, and year controls.

Interpretation. The coefficient ϕ_2 captures whether higher-wage coworkers predict the worker's future wage after moving to a new employer. It is informative about learning, but not itself structural because coworker quality is not randomly assigned.

3.2 Coworker wage regression: measuring sorting

To measure sorting, the paper estimates

$$w^*_{j^+,t+2} = \nu_0 + \nu_1 w_{i,t} + \nu_2 w^*_{j,t} + \Upsilon X_{i,j,t} + \epsilon_{i,t}, \tag{8}$$

where $w^*_{j^+,t+2}$ is the average wage of the worker's stable coworkers in the new job.

Interpretation. If $\nu_1 > 0$, then workers with higher wages tend to move to firms with higher-wage coworkers. This positive sorting must be accounted for before interpreting ϕ_2 as learning.

3.3 Production function

The model has firms with teams of up to two workers. If a firm employs workers of types k and ℓ, output is

$$f(k,\ell) = A\left(\frac{h_k^\rho + h_\ell^\rho}{2}\right)^{1/\rho}. \tag{9}$$

- If $\rho < 1$, production is supermodular: high types are more productive with high-type coworkers.
- If $\rho = 1$, production is modular.
- If $\rho > 1$, production is submodular.

Interpretation. Production supermodularity pushes toward positive sorting, because high-human-capital workers are most productive with other high-human-capital workers.

3.4 Human capital accumulation while employed

For an employed worker of type k with coworker type ℓ, the transition probability of moving up one rung is

$$g_k(k+1 \mid \ell) = \alpha_0 + \alpha_1 \frac{(h_\ell - h_k)^+}{h_N - h_1}. \tag{10}$$

The probability of staying on the same rung is $g_k(k \mid \ell) = 1 - g_k(k+1 \mid \ell)$.

Interpretation. The term α_0 is learning by doing. The term α_1 is learning from more knowledgeable coworkers. The positive-part operator implies convexity: better coworkers help, but worse coworkers do not slow the worker down.

3.5 Human capital depreciation while unemployed

For an unemployed worker,

$$g_k(k-1 \mid u) = \alpha_u, \tag{11}$$

with $g_k(k \mid u) = 1 - \alpha_u$.

Interpretation. Unemployment erodes human capital, which helps explain wage losses around EUE transitions.

3.6 Search frictions and wage timing

- Unemployed workers meet firms at rate λ_u.
- Employed workers meet poaching firms at rate λ_e.
- Exogenous separations occur with probability δ.
- The worker's bargaining power is γ.
- Wages need not immediately reflect human capital changes if the worker stays with the same employer.

Interpretation. Search frictions are crucial. They explain why the authors focus on job changers with unemployment spells: their new wages are more informative about their current human capital.

4 Identification and empirical design

4.1 What identifies the learning function

- α_1 is identified primarily by ϕ_2 in the EUE wage regression for workers who initially earn less than their coworkers.
- α_0 is identified by life-cycle wage growth.
- α_u is identified by the wage loss around EUE transitions.
- The distribution of entrants' human capital is identified by wage dispersion among young workers.

Interpretation. Higher α_1 means better coworkers cause faster catch-up. Higher α_0 means more learning by doing regardless of coworkers.

4.2 What identifies the production function

- ρ is identified by sorting patterns between workers and coworkers.
- The sorting evidence comes from the coworker wage regression and from the between-firm component of wage variance.
- Production efficiency A helps match the cross-sectional wage distribution.

Interpretation. Production complementarity is inferred from who works with whom. More positive sorting implies more complementarity in production.

4.3 Why the evidence is not a simple peer-effect regression

- Coworker wages are not randomly assigned.
- A worker's current wage and coworkers' wages are both noisy signals of the worker's latent type.
- Positive sorting means coworkers' wages can forecast future wages even without learning.
- The structural model is calibrated to the actual sorting pattern and then used to infer the causal learning function.

Interpretation. The empirical regressions motivate the learning mechanism; the structural model does the causal translation.

4.4 Validation

- The model is calibrated to 18 targets with 10 jointly chosen parameters.
- It then validates against transition rates by worker and coworker wage decile.
- The model reproduces the fact that low-wage workers have higher employment-to-unemployment rates and that job-to-job rates vary with worker-coworker combinations.

Interpretation. Matching the transition patterns supports the model's gains-from-trade structure, because hiring and separation decisions are governed by those gains.

4.5 Limitations

- The model simplifies firms as teams of at most two workers.
- Wages are noisy measures of human capital.
- The LEHD data identify coworkers within firms, but not the exact people a worker interacts with daily.
- The estimates rely on structural calibration rather than randomized assignment of coworkers.

Interpretation. These limitations matter, but the paper's strength is that it explicitly models the two main obstacles: sorting and delayed wage adjustment.

5 Tables and figures

5.1 Tables I and VII: learning regression and sorting regression

Read:

- Table I estimates the key wage-growth regression. For workers initially below their coworkers, the coworker wage coefficient is 0.123.
- For workers initially above their coworkers, the coefficient is only 0.0453.
- The interpretation is asymmetric: higher-quality coworkers strongly predict future wage growth for lower-wage workers, while lower-quality coworkers have a weaker effect on higher-wage workers.
- Table VII measures sorting. The worker's own wage predicts future coworker wages: 0.252 in the below-coworker sample and 0.172 in the above-coworker sample.
- Sorting is positive, so the wage-growth coefficients cannot be read as causal peer effects without the model.

Economic message. The reduced-form evidence points to convex learning, while the sorting regression explains why a structural model is needed.

TABLE I

INDIVIDUAL WAGE REGRESSION.

Dependent Variable	(1) Wage at $t+2$	(2) Wage at $t+2$
Sample	$w_{i,t} < w^*_{j,t}$	$w_{i,t} \geq w^*_{j,t}$
Coworker wage at t	0.123	0.0453
	(0.00327)	(0.00642)
Individual wage at t	0.513	0.773
	(0.00337)	(0.00617)
R-Squared	0.366	0.520
Round N	244,000	100,000

Note: SE clustered at SEIN level. Controls include: State, 1-digit SIC, race and gender dummies, and year fixed effects.

(a) Table I: individual wage regression.

TABLE VII

COWORKER WAGE REGRESSION.

Dependent Variable	(1) Coworker Wage at $t+2$	(2) Coworker Wage at $t+2$
Sample	$w_{i,t} < w^*_{j,t}$	$w_{i,t} \geq w^*_{j,t}$
Coworker wage at t	0.299	0.392
	(0.00369)	(0.00719)
Individual wage at t	0.252	0.172
	(0.00344)	(0.00648)
R-Squared	0.251	0.295
Round N	244,000	100,000

Note: SE clustered at SEIN level. Controls include: State, 1-digit SIC, race and gender dummies, and year fixed effects.

(b) Table VII: coworker wage regression.

5.2 Tables II and III: education and occupation heterogeneity

Read:

- Table II shows that for workers below their coworkers, the coworker effect is larger for more educated workers.
- The baseline coworker wage coefficient is 0.0585; the college interaction is 0.0992, implying a much larger effect for college workers.
- Table III shows stronger learning in abstract occupations. The baseline coefficient is 0.110; the abstract interaction is 0.0420.
- The routine interaction is negative, which means the coworker effect is lower in routine jobs.

Economic message. The heterogeneity supports the learning interpretation: coworker effects are stronger when the worker has more ability or more scope to learn.

TABLE II

INDIVIDUAL WAGE REGRESSION: EDUCATION.

Dependent Variable	(1) Wage at $t+2$	(2) Wage at $t+2$
Sample	$w_{i,t} < w^*_{j,t}$	$w_{i,t} \geq w^*_{j,t}$
Coworker wage at t	0.0585	0.0442
	(0.00913)	(0.0254)
Coworker wage at t × H	−0.00280	−0.00288
	(0.0109)	(0.0293)
Coworker wage at t × S	0.0257	−0.0220
	(0.0109)	(0.0279)
Coworker wage at t × C	0.0992	0.0387
	(0.0139)	(0.0276)
R-Squared	0.378	0.529
Round N	244,000	100,000

Note: SE clustered at SEIN level. Controls include: State, 1-digit SIC, race and gender dummies, and year fixed effects. Education classifications measured in the 2000 decennial Census. H indicates high school. S indicates some college. C indicates college or more.

(a) Table II: education heterogeneity.

TABLE III

INDIVIDUAL WAGE REGRESSION: OCCUPATION.

Dependent Variable	(1) Wage at $t+2$	(2) Wage at $t+2$
Sample	$w_{i,t} < w^*_{j,t}$	$w_{i,t} \geq w^*_{j,t}$
Coworker wage at t	0.110	0.0869
	(0.00670)	(0.0144)
Coworker wage at t × A	0.0420	0.00749
	(0.00709)	(0.0139)
Coworker wage at t × R	−0.0259	−0.0698
	(0.00667)	(0.0138)
Coworker wage at t × M	0.00243	−0.0152
	(0.00688)	(0.0136)
R-Squared	0.372	0.524
Round N	244,000	100,000

Note: SE clustered at SEIN level. Controls include: State, 1-digit SIC, race and gender dummies, and year fixed effects. Occupation classifications measured in the 2000 decennial Census. A, R, and M stand for whether the worker's occupation in the 2000 decennial census is above median in terms of abstract, routine, and manual skill requirements.

(b) Table III: occupation heterogeneity.

5.3 Tables IV and V: age and coworker wage distribution

Read:

- Table IV shows that for workers below coworkers, the coworker wage coefficient declines with age.
- The youngest group has a coefficient of 0.148, while the tenth age decile has an interaction of -0.0466, implying a much smaller effect for older workers.
- Table V adds dispersion, minimum, and maximum coworker wages.
- The average coworker wage coefficient rises when dispersion controls are added, while the standard deviation coefficient is negative.
- This is consistent with measurement error: average coworker wage is a noisier proxy for relevant peer human capital when coworker dispersion is high.

Economic message. Younger workers and workers in less dispersed learning environments are more responsive to coworker quality.

TABLE IV

INDIVIDUAL WAGE REGRESSION: AGE.

Dependent Variable	(1) Wage at $t+2$	(2) Wage at $t+2$
Sample	$w_{i,t} < w^*_{j,t}$	$w_{i,t} \geq w^*_{j,t}$
Coworker wage at t	0.148	0.0596
	(0.00820)	(0.0320)
Coworker wage at t × Age decile 2	−0.0197	0.0189
	(0.0123)	(0.0415)
Coworker wage at t × Age decile 3	−0.0279	−0.0154
	(0.0122)	(0.0378)
Coworker wage at t × Age decile 4	−0.0259	−0.0344
	(0.0133)	(0.0385)
Coworker wage at t × Age decile 5	−0.0351	−0.0199
	(0.0125)	(0.0358)
Coworker wage at t × Age decile 6	−0.0428	0.00314
	(0.0133)	(0.0376)
Coworker wage at t × Age decile 7	−0.0297	0.00709
	(0.0138)	(0.0365)
Coworker wage at t × Age decile 8	−0.0395	−0.00877
	(0.0128)	(0.0361)
Coworker wage at t × Age decile 9	−0.0397	−0.0194
	(0.0131)	(0.0361)
Coworker wage at t × Age decile 10	−0.0466	−0.0816
	(0.0144)	(0.0383)
R-Squared	0.372	0.525
Round N	244,000	100,000

(a) Table IV: age heterogeneity.

TABLE V

INDIVIDUAL WAGE REGRESSION: OTHER MOMENTS OF COWORKERS' WAGES.

Dependent Variable	(1) Wage at $t+2$	(2) Wage at $t+2$	(3) Wage at $t+2$	(4) Wage at $t+2$
Sample	$w_{i,t} < w^*_{j,t}$	$w_{i,t} < w^*_{j,t}$	$w_{i,t} \geq w^*_{j,t}$	$w_{i,t} \geq w^*_{j,t}$
Coworker wage at t	0.191	0.226	0.0537	0.0441
	(0.00481)	(0.00553)	(0.00646)	(0.00891)
Individual wage at t	0.496	0.513	0.801	0.789
	(0.00351)	(0.00345)	(0.00682)	(0.00667)
Std of coworker wages, t	−0.0410		−0.0306	
	(0.00206)		(0.00309)	
Min of coworker wage at t		−0.0222		0.0306
		(0.00276)		(0.00415)
Max of coworker wage at t		−0.0684		−0.0287
		(0.00264)		(0.00530)
R-Squared	0.367	0.368	0.521	0.521
Round N	244,000	244,000	100,000	100,000

Note: SE clustered at SEIN level. Controls include: State, 1-digit SIC, race and gender dummies, and year fixed effects. Minimum, maximum, and standard deviation of coworkers wages are computed among stable coworkers at t.

(b) Table V: other coworker wage moments.

5.4 Table VI: stayers versus movers

Read:

- Table VI repeats the wage regression for workers who stay at the same firm from year t to $t+2$.

- Coworker wage coefficients are lower than in Table I: 0.0600 for workers below coworkers and 0.0301 for workers above coworkers.
- The R^2 is much higher for stayers, reflecting persistent wages within the same employment relationship.

Economic message. The lower coefficients for stayers reinforce the paper's search-friction logic: wages at the same firm do not immediately reveal new human capital.

TABLE VI

INDIVIDUAL WAGE REGRESSION: STAYERS.

Dependent Variable	(1) Wage at $t+2$	(2) Wage at $t+2$
Sample	$w_{i,t} < w^*_{j,t}$	$w_{i,t} > w^*_{j,t}$
Coworker wage at t	0.0600	0.0301
	(0.000418)	(0.000423)
Individual wage at t	0.908	0.950
	(0.000438)	(0.000391)
R-Squared	0.831	0.872
Round N	6.732e+06	5.392e+06

Note: SE clustered at SEIN level. Controls include: State, 1-digit SIC, race and gender dummies, and year fixed effects.

Table VI: wage regression for stayers.

5.5 Tables VIII and IX plus Figures 1 and 2: calibration fit and parameter values

Read:

- Table VIII shows that the model matches the key learning targets closely: $\phi_2 = 0.12$ in the data and 0.12 in the model for workers below coworkers.
- The model also matches life-cycle wage growth, transition rates, and the flow value of unemployment reasonably well.
- Table IX reports the calibrated values: $\alpha_0 = 0.001$, $\alpha_1 = 0.020$, $\alpha_u = 0.016$, and $\rho = 0.810$.
- α_1 is much larger than α_0, implying that learning from better coworkers is quantitatively important.
- Figure 1 validates the model against EU and EE rates by worker and coworker wage decile.
- Figure 2 shows that the model captures lifetime earnings growth reasonably well over the middle of the lifetime earnings distribution.

Economic message. The model is not just fitting wage regressions. It also produces realistic transition patterns and life-cycle wage growth, supporting the calibration.

TABLE VIII

CALIBRATION TARGETS AND MODEL FIT.

Target	Source	Data	Model
ϕ_1, $w_{it} \leq w^*_{jt}$	LEHD	0.51	0.52
ϕ_1, $w_{it} > w^*_{jt}$	LEHD	0.77	0.78
ϕ_2, $w_{it} \leq w^*_{jt}$	LEHD	0.12	0.12
ϕ_2, $w_{it} > w^*_{jt}$	LEHD	0.05	0.03
Between-firm wage variance	Song et al. (2019)	0.40	0.44
υ_1, $w_{it} \leq w^*_{jt}$	LEHD	0.25	0.32
υ_1, $w_{it} > w^*_{jt}$	LEHD	0.17	0.22
υ_2, $w_{it} \leq w^*_{jt}$	LEHD	0.30	0.12
υ_2, $w_{it} > w^*_{jt}$	LEHD	0.39	0.02
EUE average wage loss	LEHD	−0.18	−0.22
54-to-24 y.o. wage ratio	CPS	1.88	1.93
Mean wage growth	CPS	0.02	0.02
p90/p10 wage ratio	CPS	4.23	3.50
p90/p10 wage ratio 24 y.o.	CPS	2.77	2.07
UE Rate	CPS	0.22	0.24
EE Rate	CPS	0.02	0.01
EU Rate	CPS	0.01	0.01
Flow unemployment value	Hall and Milgrom (2008)	0.71	0.73

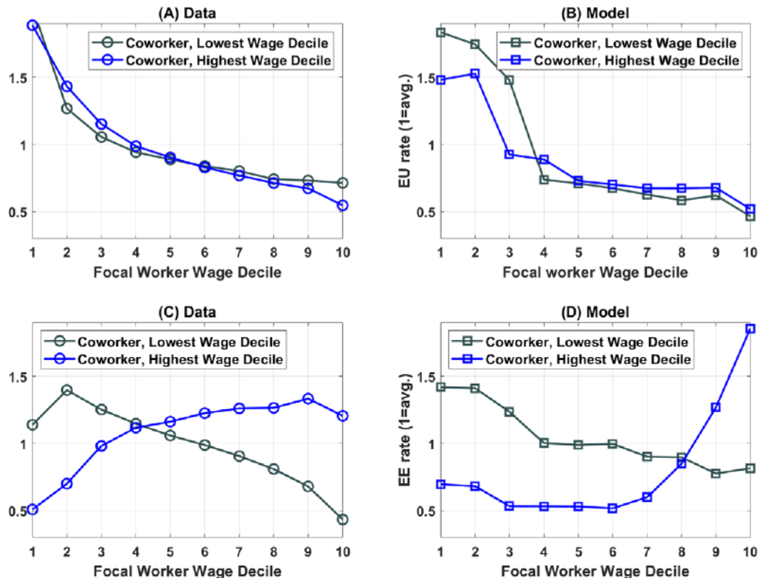
(a) Table VIII: calibration targets and model fit.

TABLE IX

CALIBRATED PARAMETER VALUES.

Parameter	Description	Value
α_0	Learning by doing	0.001
α_1	Learning from coworkers	0.020
α_u	Human capital depreciation	0.016
ρ	Production complementarity	0.810
A	Production efficiency	2.194
χ	Entrant distribution	2.623
λ_u	Meeting rate, unemployment	0.340
λ_e	Meeting rate, employed	0.238
δ	Separation rate	0.009
b	Flow value of unemployment	0.976
β	Discount factor	0.992
σ	Exit rate	0.002
γ	Bargaining power worker	0.5
h_N	Human capital ladder	5.0

(b) Table IX: calibrated parameter values.



(a) Figure 1: transition-rate validation.

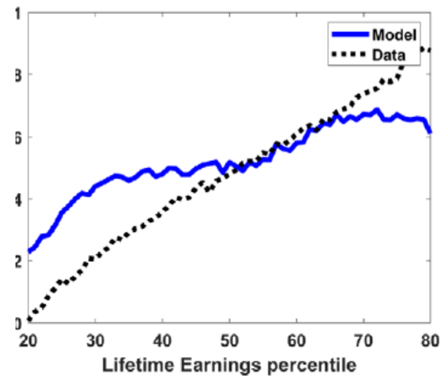


FIGURE 2.—Lifetime Earnings growth.

(b) Figure 2: lifetime earnings growth.

5.6 Figures 3 and 4: policy functions and worker-coworker distributions

Read:

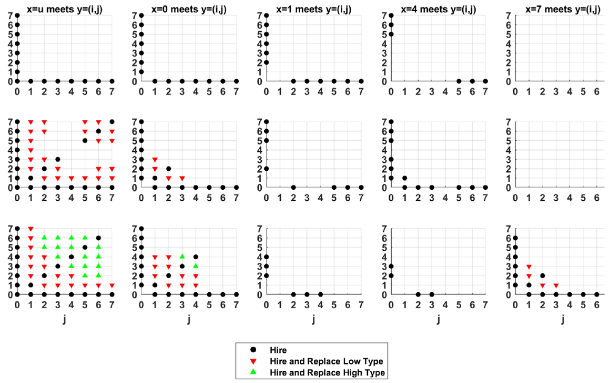
- Figure 3 displays equilibrium hiring and replacement policies across worker types and firm states.
- The policy functions show that firms' decisions depend strongly on the type of the worker already in the firm and the type of the worker being met.
- Figure 4 shows the equilibrium and counterfactual distributions of worker-coworker pairs.
- The baseline has full support: the model is neither pure positive assortative matching nor pure negative assortative matching.
- The no-learning-from-coworkers counterfactual has more density along the off-diagonal than the baseline, meaning learning from coworkers makes equilibrium sorting less positive.
- The high-supermodularity counterfactual increases segregation: workers are more likely to match with similar coworkers.

Economic message. Sorting is determined jointly by current production and future learning. Stronger production complementarity raises segregation, but learning from better coworkers creates value in mixed teams.

5.7 Figures 5-7: transitions, life-cycle human capital, and type distribution

Read:

- Figure 5 shows that unemployment is much higher for low-human-capital workers: around 10 percent at the bottom rung and around 2 percent at the top rung.
- Higher-human-capital workers have higher UE rates and lower EU rates.
- Figure 6 shows human capital and wages over the life cycle. Wages grow faster than human capital because workers also move to better firms and capture a rising share of match value.
- Figure 7 compares entering workers, active workers, and counterfactual active-worker distributions.



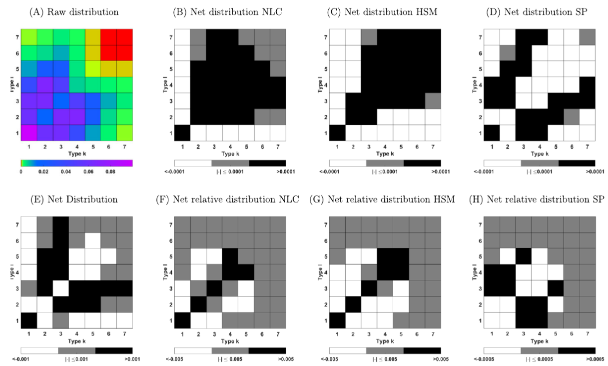
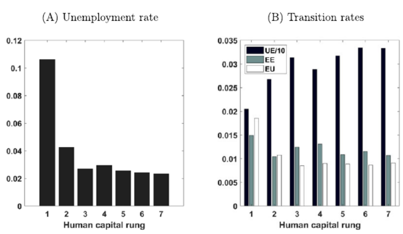


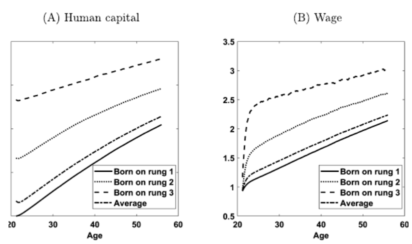
FIGURE 4.—Net and relative distributions of workers and coworkers. Notes: Panel A is the baseline distribution of workers and coworkers. Panels B through E lot the difference between the fraction of teams with a worker of type k and a worker of type l net of the fraction of teams with a worker of type k and a worker of pe l if teams were formed at random in the NLC counterfactual and in the baseline. Panels F through H plot the net distribution after subtracting the benchmark odel's net distribution.

- Active workers have much more human capital than entrants. Without learning from coworkers, the active-worker distribution shifts lower.
- The paper reports that active workers' human capital is 70 percent higher than entrants' in the baseline but only 21 percent higher in the no-learning-from-coworkers model.

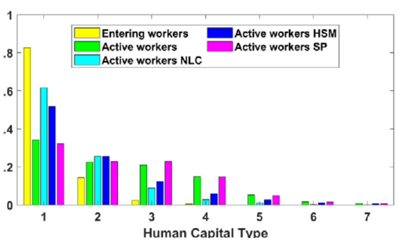
Economic message. Learning on the job is quantitatively large, and learning from coworkers is the dominant part of that on-the-job accumulation.



(a) Figure 5: unemployment and transition rates.



(b) Figure 6: human capital and wages.



(c) Figure 7: type distribution.

6 Short summary

- The paper studies whether workers learn from coworkers and how this affects sorting, human capital, and output.
- The authors use LEHD matched employer-employee data linked to Census information.
- The main empirical sample is workers who move from employment to unemployment to employment at a different firm.
- For workers initially earning less than coworkers, higher coworker wages predict substantially higher future wages.
- For workers initially earning more than coworkers, lower coworker quality has a much weaker effect.
- Education, abstract occupations, and youth make the coworker effect stronger.
- Stayers have smaller coworker coefficients, supporting the search-friction assumption that wages do not immediately reflect human capital growth.
- Sorting is positive: higher-wage workers tend to move to firms with higher-wage coworkers.
- The production function is estimated to be mildly supermodular.
- The human capital accumulation function is estimated to be convex in coworker human capital.
- Learning from coworkers accounts for roughly two thirds of on-the-job human capital accumulation.
- No-learning-from-coworkers reduces the aggregate stock of active-worker human capital by 29 percent and output by 27 percent.
- Higher production supermodularity raises segregation and reduces learning opportunities for low-human-capital workers.
- The planner would choose less positive sorting than the decentralized equilibrium, but the quantified welfare loss is modest.

7 Conclusion

7.1 Core conclusion

Herkenhoff, Lise, Menzio, and Phillips show that coworker quality is a quantitatively important input into human capital accumulation. The key asymmetry is that workers learn from more knowledgeable coworkers, while less knowledgeable coworkers do not significantly slow down more knowledgeable workers.

7.2 Economic intuition

Workers and coworkers are linked through two channels. The production channel rewards sorting similar types together when production is supermodular. The learning channel rewards mixing low-human-capital workers with higher-human-capital coworkers. Equilibrium sorting balances these forces under search frictions and bargaining wedges.

7.3 Takeaway

Workplace sorting affects both current output and future human capital. Policies or technologies that increase segregation can have dynamic costs by reducing knowledge diffusion to low-human-capital workers. The main lesson is that coworkers are not just peer inputs into current production; they are also teachers who shape the future distribution of skill.